Introduction to Data Analytics 2

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Machine: Siddheshs-MacBook-Pro.local - Darwin 25.1.0

Part 1: Experiment Design

Title: Consumer Pseudo-Showrooming and Omni-Channel Product Placement Strategies

Abstract: Recent advances in information technologies (IT) have powered the merger of online and offline retail channels into one single platform. Modern consumers frequently switch between online and offline channels when they navigate through various stages of the decision journey, motivating multichannel sellers to develop omni-channel strategies that optimize their overall profit. This study examines consumers' cross-channel search behavior of "pseudo-showrooming," or the consumer behavior of inspecting one product at a seller's physical store before buying a related but different product at the same seller's online store, and investigates how such consumer behavior allows a multichannel seller to achieve better coordination between its online and offline arms through optimal product placement strategies.

Participants in the study were grouped into the following categories:

- Where_bought: Where they ended up purchasing an item: bought at the store, bought online.
- Who_bought: If they bought from the same or a different retailer.

Each participant was then measured on:

- Money: how much money they spent in dollars on the product.
- Time: how much time (in minutes) they spent looking at the product online.

1) What would be one possible null hypothesis based on this study?

One possible null hypothesis based on this study could be: "There is no significant difference in the amount of money spent on products between consumers who purchase from the same retailer versus those who purchase from a different retailer, and there is no difference between online and in-store purchases."

2) What would be one possible alternative hypothesis based on this study?

The alternative hypothesis based on this study could be: "There is a significant difference in the amount of money spent on products based on where consumers purchase (online vs. store) and whether they purchase from the same or different retailer."

3) Who are they sampling in this study?

The sample consists of consumers who engage in cross-channel shopping behavior when they inspect the products in physical store and potentially purchase either online or in-store, from either the same retailer or different retailer.

4) Who is the intended population in this study?

The intended population is all consumers who browse products across different retail channels before making decisions.

5) Give an example of type 1 error based on this study (do not just define, explain in context how it might have happened here).

A Type 1 error (False Positive) would occur if researchers concluded that purchasing channel (online or store) significantly affects money spent when, in reality, it doesn't.

An example could be when the study might find that the customers who buy online from same retailer spend more money that those who buy from the store, leading the retailer to invest more in online promotions and inventory. However, if the difference was due to random chance or sampling issues, the retailer would waste resources on this strategy which was based on a false finding.

6) Give an example of type 2 error based on this study (do not just define, explain in context how it might have happened here).

A Type 2 error (False Negative) would occur if the researchers failed to detect a real difference in spending behavior conditions.

An example could be when the study might find that there is no significant difference in money spent between customers who pseudo-showroom at the same retailer versus those who switch to a different retailer. Based on this, the company doesn't implement strategy to retain people who visit their showrooms. However, if the study failed to detect the fact that same-retail shoppers do spend significantly more, the company misses an opportunity to optimize their omni-channel strategy and loses potential revenue.

Part 2: Use the 04_data.csv to complete this portion.

I have used `read_csv()` to load the dataset as `df` variable

```
df <- read.csv("04_data.csv")
head(df)
```

```
##     id where_bought who_bought    money      time
## 1 S001       online  different 26.63386  9.134374
## 2 S002       online  different 26.69806 10.711019
## 3 S003       online  different 21.79154  7.549178
## 4 S004       online  different 24.31975 11.485499
## 5 S005       online  different 26.86524  6.068067
## 6 S006       online  different 28.32115 10.796021
```

1) For each IV list the levels (next to a, b):

I have used `unique()` to get all unique values for `where_bought` and `who_bought` from the dataset

```
a.  Where bought:

We get "online" and "store" as different levels for Where bought
```

```
where_bought_levels <- unique(df$where_bought)
where_bought_levels
```

```
## [1] "online" "store"
```

```
b.  Who bought:

We get "different" and "same" as different levels for Who bought
```

```
who_bought_levels <- unique(df$who_bought)
who_bought_levels
```

```
## [1] "different" "same"
```

2) What are the conditions in this experiment?

The conditions in this experiment are formed by the combinations of levels of independent variables. So the conditions are:

- online and different
- online and same
- store and different
- store and same

3) For each condition list the means, standard deviations, and standard error for the conditions for time and money spent. Please note that means you should have several sets of M, SD, and SE. Be sure you name the sets of means, sd, and se different things so you can use them later.

I have imported `dplyr` package for performing data manipulation. Using the function `group_by()` and `summarise()` on `df`, I calculated mean, sd and se for time and money based on the conditions that we found above.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##     filter, lag
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# group statistics for time
stats_time <- df %>%
  group_by(where_bought, who_bought) %>%
  summarise(
    mean_time = mean(time, na.rm = TRUE),
    sd_time = sd(time, na.rm = TRUE),
    n = n(),
    se_time = sd(time, na.rm = TRUE) / sqrt(n),
    .groups = 'drop'
  )

# group statistics for money
stats_money <- df %>%
  group_by(where_bought, who_bought) %>%
  summarise(
    mean_money = mean(money, na.rm = TRUE),
    sd_money = sd(money, na.rm = TRUE),
    n = n(),
    se_money = sd(money, na.rm = TRUE) / sqrt(n),
    .groups = 'drop'
  )

print(stats_time)
```

```
## # A tibble: 4 x 6
##   where_bought who_bought mean_time sd_time     n se_time
##   <chr>        <chr>          <dbl>   <dbl> <int>   <dbl>
```

```
## 1 online       different       10.0     2.75     50    0.389
## 2 online       same            10.5     3.03     50    0.428
## 3 store        different       15.3     2.66     50    0.376
## 4 store        same            20.1     2.91     50    0.412
```

```
print(stats_money)
```

```
## # A tibble: 4 x 6
##   where_bought who_bought mean_money sd_money     n se_money
##   <chr>        <chr>           <dbl>    <dbl> <int>    <dbl>
## 1 online       different        25.4     2.81    50    0.397
## 2 online       same             25.1     3.26    50    0.461
## 3 store        different        34.7     3.05    50    0.432
## 4 store        same             34.9     2.74    50    0.387
```

4) Which condition appears to have the best model fit using the mean as the model (i.e. smallest error) for time?

I have used `filter()` with a condition where the SE of time is minimum and based on the output, we see that store-different appears to have best model fit.

```
best_fit_time <- stats_time %>% filter(se_time == min(se_time))
best_fit_time
```

```
## # A tibble: 1 x 6
##   where_bought who_bought mean_time sd_time     n se_time
##   <chr>        <chr>          <dbl>   <dbl> <int>   <dbl>
## 1 store        different       15.3    2.66    50   0.376
```

5) What are the df for each condition?

I have used `group_by()` and `summarise()` on `df` to get df for each out conditions i.e. n-1 which comes to 49 for all conditions.

```
condition_df <- df %>%
  group_by(where_bought, who_bought) %>%
  summarise(df = n() - 1, .groups = 'drop')
condition_df
```

```
## # A tibble: 4 x 3
##   where_bought who_bought    df
##   <chr>        <chr>      <dbl>
## 1 online       different     49
## 2 online       same          49
## 3 store        different     49
## 4 store        same          49
```

6) What is the confidence interval (95%) for the means?

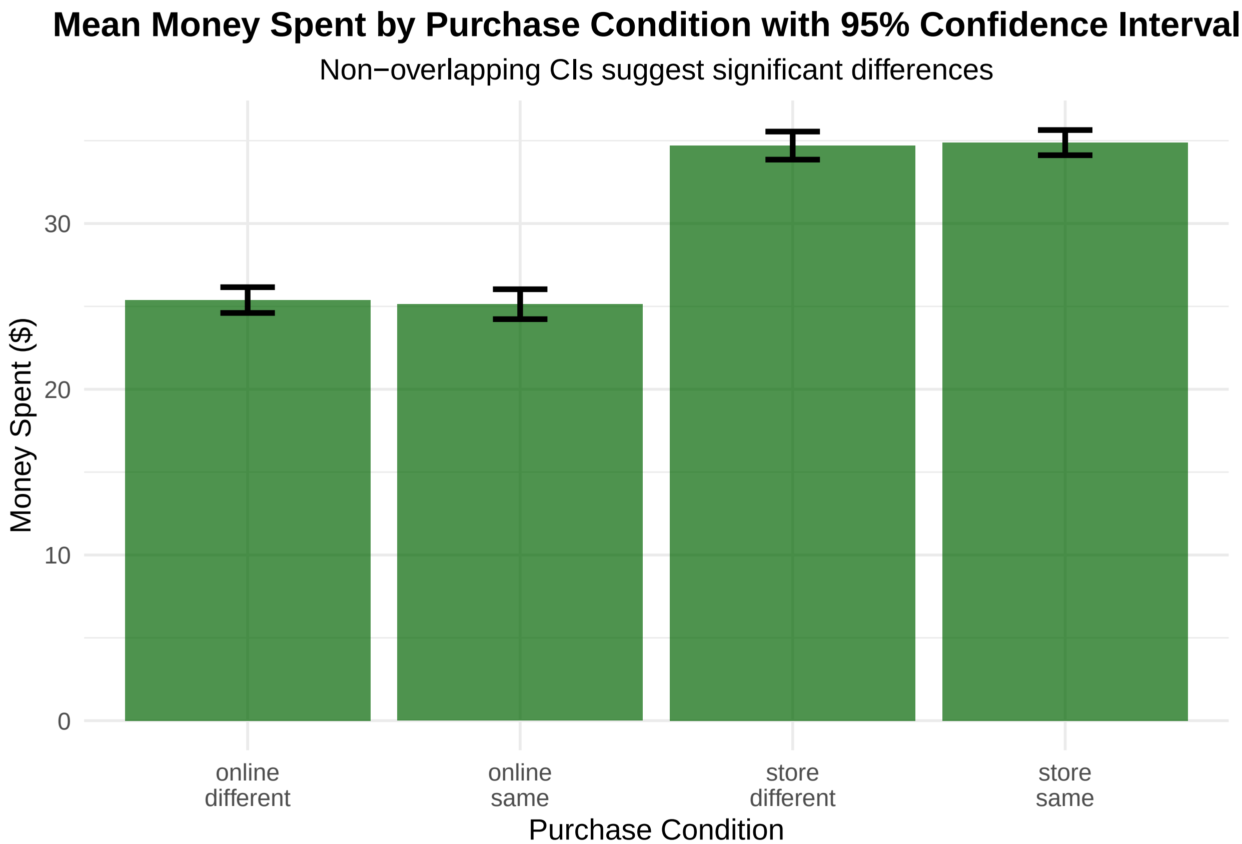
I have used the formula `sample_mean +- z_95 * standard_error` to calculate lower and upper limit of Confidence Intervals of money and time. I have then combines the conditions `where_bought` and `who_bought` and used `ggplot2` library to plot the respective Confidence Intervals.

```
library(ggplot2)

z_95 <- 1.96

# Money
confidence_interval_money <- stats_money %>%
  mutate(
```

```r
    ci_lower_money = mean_money - z_95 * se_money,
    ci_upper_money = mean_money + z_95 * se_money
  )

print(confidence_interval_money)
```

```
## # A tibble: 4 x 8
##   where_bought who_bought mean_money sd_money     n se_money ci_lower_money
##   <chr>        <chr>           <dbl>    <dbl> <int>    <dbl>          <dbl>
## 1 online       different        25.4     2.81    50    0.397           24.6
## 2 online       same             25.1     3.26    50    0.461           24.2
## 3 store        different        34.7     3.05    50    0.432           33.9
## 4 store        same             34.9     2.74    50    0.387           34.1
## # i 1 more variable: ci_upper_money <dbl>
```

```r
ci_money_plot <- confidence_interval_money %>%
  mutate(condition = paste(where_bought, who_bought, sep = "\n"))

ggplot(ci_money_plot, aes(x = condition, y = mean_money)) +
  geom_col(fill = "darkgreen", alpha = 0.7) +
  geom_errorbar(aes(ymin = ci_lower_money, ymax = ci_upper_money),
                width = 0.2, linewidth = 1) +
  labs(title = "Mean Money Spent by Purchase Condition with 95% Confidence Intervals",
       subtitle = "Non-overlapping CIs suggest significant differences",
       x = "Purchase Condition",
       y = "Money Spent ($)") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(hjust = 0.5))
```

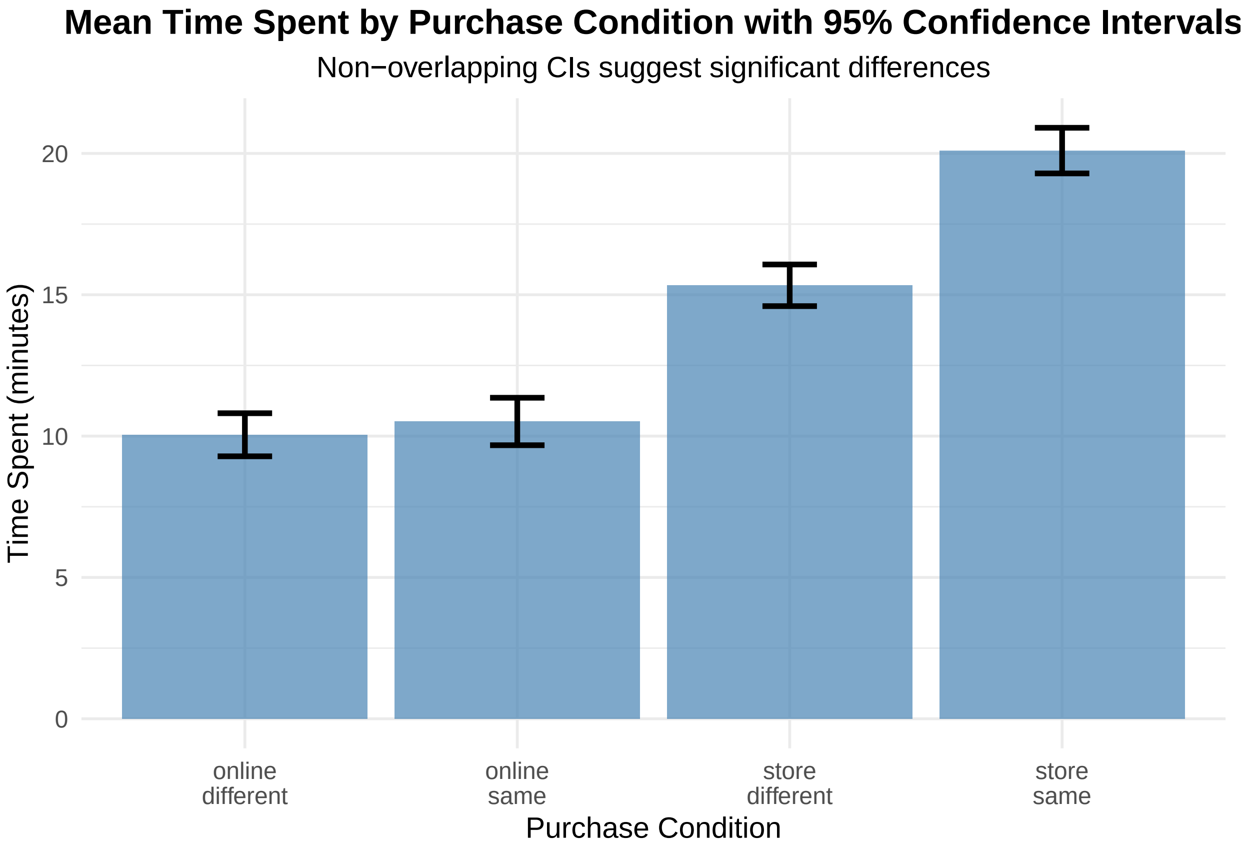



```
# Time
confidence_interval_time <- stats_time %>%
  mutate(
    ci_lower_time = mean_time - z_95 * se_time,
    ci_upper_time = mean_time + z_95 * se_time
  )

print(confidence_interval_time)
```

```
## # A tibble: 4 x 8
##   where_bought who_bought mean_time sd_time     n se_time ci_lower_time
##   <chr>        <chr>          <dbl>   <dbl> <int>   <dbl>         <dbl>
## 1 online       different       10.0    2.75    50   0.389          9.29
## 2 online       same            10.5    3.03    50   0.428          9.68
## 3 store        different       15.3    2.66    50   0.376         14.6
## 4 store        same            20.1    2.91    50   0.412         19.3
## # i 1 more variable: ci_upper_time <dbl>
```

```
ci_time_plot <- confidence_interval_time %>%
  mutate(condition = paste(where_bought, who_bought, sep = "\n"))

ggplot(ci_time_plot, aes(x = condition, y = mean_time)) +
  geom_col(fill = "steelblue", alpha = 0.7) +
  geom_errorbar(aes(ymin = ci_lower_time, ymax = ci_upper_time),
                width = 0.2, linewidth = 1) +
  labs(title = "Mean Time Spent by Purchase Condition with 95% Confidence Intervals",
       subtitle = "Non-overlapping CIs suggest significant differences",
       x = "Purchase Condition",
```

```r
      y = "Time Spent (minutes)") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(hjust = 0.5))
```



7) Use the MOTE library to calculate the effect size for the difference between money spent for the following comparisons (that means you'll have to do this twice):

I have used `d.int.t()` from MOTE library to calculate effect size for the difference between money spent when:

1. Store versus online when bought at the same retailer
2. Store versus online when bought at a different retailer

I have used the value of alpha = 0.05. We find out that d = -3.240354 for Same Retailer and d = -3.180373 for Different Retailer.

```r
library(MOTE)

alpha <- 0.05 # significance level

# Store versus online when bought at the same retailer
effect_money_same <- d.ind.t(
  m1 = stats_money$mean_money[stats_money$where_bought == "online" &
                                stats_money$who_bought == "same"],
  m2 = stats_money$mean_money[stats_money$where_bought == "store" &
                                stats_money$who_bought == "same"],
  sd1 = stats_money$sd_money[stats_money$where_bought == "online" &
```

```
                               stats_money$who_bought == "same"],
  sd2 = stats_money$sd_money[stats_money$where_bought == "store" &
                               stats_money$who_bought == "same"],
  n1 = stats_money$n[stats_money$where_bought == "online" &
                       stats_money$who_bought == "same"],
  n2 = stats_money$n[stats_money$where_bought == "store" &
                       stats_money$who_bought == "same"],
  a = alpha
)

print("\nFor Same Retailer: \n")
```

```
## [1] "\nFor Same Retailer: \n"
```

```
print(effect_money_same)
```

```
## $d
## [1] -3.240354
##
## $dlow
## [1] -3.83566
##
## $dhigh
## [1] -2.637545
##
## $M1
## [1] 25.13133
##
## $sd1
## [1] 3.256608
##
## $se1
## [1] 0.460554
##
## $M1low
## [1] 24.20581
##
## $M1high
## [1] 26.05685
##
## $M2
## [1] 34.88091
##
## $sd2
## [1] 2.738662
##
## $se2
## [1] 0.3873052
##
## $M2low
## [1] 34.10259
##
## $M2high
## [1] 35.65923
```

```
##
## $spooled
## [1] 3.008801
##
## $sepooled
## [1] 0.6017602
##
## $n1
## [1] 50
##
## $n2
## [1] 50
##
## $df
## [1] 98
##
## $t
## [1] -16.20177
##
## $p
## [1] 1.796274e-29
##
## $estimate
## [1] "$d_s$ = -3.24, 95\\% CI [-3.84, -2.64]"
##
## $statistic
## [1] "$t$(98) = -16.20, $p$ < .001"
```

```r
# Store versus online when bought at a different retailer
effect_money_different <- d.ind.t(
  m1 = stats_money$mean_money[stats_money$where_bought == "online" &
                                stats_money$who_bought == "different"],
  m2 = stats_money$mean_money[stats_money$where_bought == "store" &
                                stats_money$who_bought == "different"],
  sd1 = stats_money$sd_money[stats_money$where_bought == "online" &
                               stats_money$who_bought == "different"],
  sd2 = stats_money$sd_money[stats_money$where_bought == "store" &
                               stats_money$who_bought == "different"],
  n1 = stats_money$n[stats_money$where_bought == "online" &
                       stats_money$who_bought == "different"],
  n2 = stats_money$n[stats_money$where_bought == "store" &
                       stats_money$who_bought == "different"],
  a = alpha
)

print("\nFor Different Retailer: \n")
```

```
## [1] "\nFor Different Retailer: \n"
```

```r
print(effect_money_different)
```

```
## $d
## [1] -3.180373
##
## $dlow
```

```
## [1] -3.769363
##
## $dhigh
## [1] -2.583877
##
## $M1
## [1] 25.38076
##
## $sd1
## [1] 2.805442
##
## $se1
## [1] 0.3967494
##
## $M1low
## [1] 24.58346
##
## $M1high
## [1] 26.17806
##
## $M2
## [1] 34.70278
##
## $sd2
## [1] 3.051608
##
## $se2
## [1] 0.4315625
##
## $M2low
## [1] 33.83553
##
## $M2high
## [1] 35.57004
##
## $spooled
## [1] 2.93111
##
## $sepooled
## [1] 0.586222
##
## $n1
## [1] 50
##
## $n2
## [1] 50
##
## $df
## [1] 98
##
## $t
## [1] -15.90187
##
## $p
```

```
## [1] 6.801843e-29
##
## $estimate
## [1] "$d_s$ = -3.18, 95\\% CI [-3.77, -2.58]"
##
## $statistic
## [1] "$t$(98) = -15.90, $p$ < .001"
```

8) What can you determine about the effect size in the experiment - is it small, medium or large?

For Cohen's d benchmarks: Small effect: |d| ~ 0.2 Medium effect: |d| ~ 0.5 Large effect: |d| >= 0.8

Since the absolute value of d >= 0.8 in both cases, based on Cohen's benchmarks, both the cases have large effect.

9) How many people did we need in the study for each comparison?

```r
library(pwr)

# Assume the required variables
power <- 0.80
alpha <- 0.05

# For different retailer
power_different <- pwr.t.test(
  d = effect_money_different$d,  # expected effect size (medium)
  power = power,                # desired statistical power
  sig.level = alpha,            # prob of false positive
  type = "two.sample",
  alternative = "two.sided"
)

cat("\nDifferent Retailer Comparison\n")
```

```
##
## Different Retailer Comparison
```

```r
print(power_different)
```

```
##
##      Two-sample t test power calculation
##
##               n = 2.901572
##               d = 3.180373
##       sig.level = 0.05
##           power = 0.8
##     alternative = two.sided
##
## NOTE: n is number in *each* group
```

```r
cat("Sample size needed per group:", ceiling(power_different$n), "\n")
```

```
## Sample size needed per group: 3
```

```r
# For same retailer
power_same <- pwr.t.test(
  d = effect_money_same$d,  # expected effect size (medium)
  power = power,                # desired statistical power
  sig.level = alpha,            # prob of false positive
```

```r
  type = "two.sample",
  alternative = "two.sided"
)

cat("\nSame Retailer Comparison\n")
```

```
##
## Same Retailer Comparison
```

```r
print(power_same)
```

```
##
##      Two-sample t test power calculation
##
##               n = 2.852122
##               d = 3.240354
##       sig.level = 0.05
##           power = 0.8
##     alternative = two.sided
##
## NOTE: n is number in *each* group
```

```r
cat("Sample size needed per group:", ceiling(power_same$n), "\n")
```

```
## Sample size needed per group: 3
```

I have used `pwr.t.test` to calculate the required sample size for Cohen's d that we calculated above. Based on the test, we should have n = 3 people in each group to get standardized difference between two group means (Cohen's d)